Business Requirements

Functional Requirements

- Collect soil and environmental parameters from users (Nitrogen, Phosphorus, Potassium, temperature, humidity, pH, rainfall).
- Analyze the agricultural dataset using data visualization techniques.
- Perform data preprocessing, including handling missing values and outliers.
- Implement K-Means Clustering to identify similar agricultural conditions.
- Train a Logistic Regression model for crop recommendation.
- Recommend the most suitable crop based on user inputs.
- Display prediction results through an intuitive web interface.

Non-Functional Requirements

- Responsive and user-friendly interface.
- High prediction accuracy.
- Fast response time.
- Secure data handling.
- Scalable architecture for future expansion.
- Easy integration with cloud services and IoT devices.
- The system should remain available with minimal downtime

Business Goals

- Improve crop productivity.
- Reduce crop failure.
- Optimize fertilizer and water usage.
- Increase farmers' profitability.
- Encourage sustainable farming practices.